

CO1-AT2

Agile Sprint Planning and User Story Workshop

Intelligent Electric Vehicle Charging Network Management System



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Course: CSA1014 Software Engineering



Industry Context



Rapid EV growth — Electric-vehicle adoption is rising sharply, driven by cost, policy and sustainability goals.



Charging demand — Public and workplace charging demand is outpacing the rollout of new charging infrastructure.



Long waiting times — Drivers queue at busy stations with no reliable way to know how long they will wait.



Poor charger visibility — Real-time availability, location and status of chargers are rarely visible to drivers.



Need for smart management — AI and IoT can monitor, predict and optimise charging across the whole network.



Smart EV Charging Station



Connected EVs



AI prediction



IoT sensors

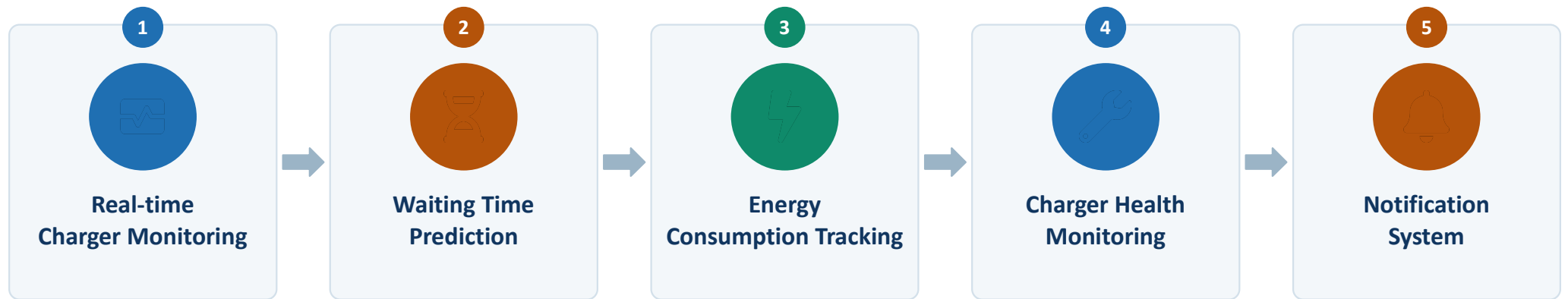


Clean energy

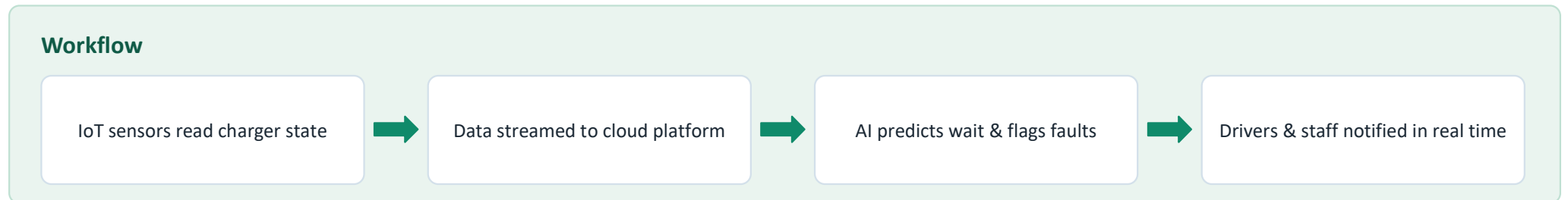


Problem Statement

The system must address five interconnected capabilities, each feeding the next:



Workflow





Project Vision & Objectives

Vision To deliver an intelligent, connected EV-charging network that gives drivers real-time visibility, minimises waiting, and helps operators optimise energy and maintenance across every station.



Monitor charging stations

Track live status, availability and usage of every charger in the network.



Predict waiting time

Use AI to forecast queue and charge-completion times for each station.



Optimise charger allocation

Guide drivers to the best available charger to balance network load.



Generate energy reports

Produce consumption and efficiency reports for administrators.



Improve customer satisfaction

Reduce uncertainty and waiting to deliver a smoother charging experience.



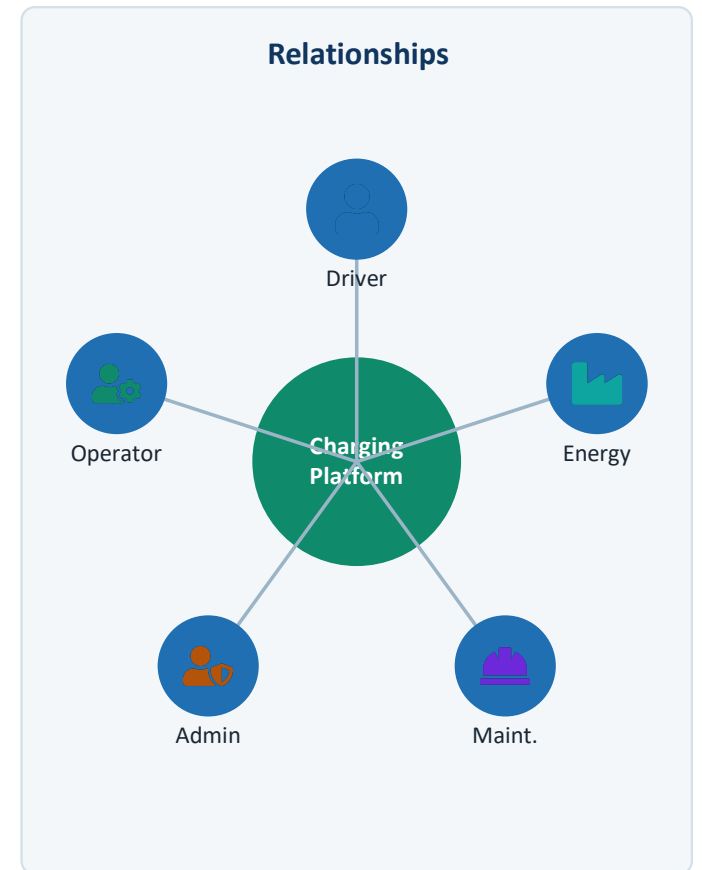
Smart, sustainable charging

Every objective feeds one goal: an efficient, greener charging ecosystem.



Stakeholder Analysis

Stakeholder	Role	Responsibilities	Challenges
EV Driver	End user	Find, reserve and use chargers; pay for sessions.	Long waits and uncertain charger availability.
Station Operator	Service provider	Run stations; ensure chargers are online and available.	Balancing demand and uptime across sites.
Administrator	System manager	Oversee network, users and energy reports.	Turning raw data into actionable decisions.
Maintenance Team	Support	Diagnose and repair faulty chargers quickly.	Detecting faults early to cut downtime.
Energy Provider	Supplier	Supply and bill electricity to the network.	Managing peak load and grid stability.



Epics — Agile Roadmap





User Stories



As a EV Driver

I want to **register**

so that I can access charging services.



As a EV Driver

I want to **view nearby charging stations**

so that I can choose the nearest charger.



As a EV Driver

I want to **reserve a charger**

so that I don't have to wait.



As a Administrator

I want to **view energy reports**

so that I can monitor electricity usage.



As a Maintenance Engineer

I want to **receive charger fault alerts**

so that I can repair faulty chargers quickly.



Acceptance Criteria

Given–When–Then — Given a context, When an action occurs, Then the system produces a defined, testable outcome.

Scenario	Given	When	Then
Registration	A new user on the signup page	they submit valid details	the account is created and confirmed.
Login	A registered user	they enter correct credentials	they are granted access to the dashboard.
Charger Availability	A driver viewing the map	they open a station	live availability of each charger is shown.
Reservation	An available charger	the driver reserves it	the charger is held for that driver.
Waiting Time Prediction	A busy station	the driver requests an estimate	a predicted waiting time is displayed.



Product Backlog

Priority	Epic	User Story	Business Value	Story Points
High	User Management	Register and log in to the platform	Essential access to all services	5
High	Station Monitoring	View live charger availability	Core value — drivers find chargers	5
High	Session Management	Reserve a charger in advance	Directly cuts waiting time	8
Medium	Waiting Time Prediction	See predicted waiting time	Improves planning and trust	8
Medium	Notification System	Get availability and fault alerts	Keeps users and staff informed	5
Low	Energy Reporting	Generate energy usage reports	Supports admin optimisation	5
Low	Maintenance Management	Track and resolve charger faults	Reduces long-term downtime	5

Priority: High Medium Low



Sprint Backlog — Scrum Board

Sprint Duration: 2 Weeks

Sprint Goal: Develop the core charging management features.

To Do

Station Monitoring

Core feature

Charger Availability

Core feature

In Progress

Login

Core feature

Done

Registration

Core feature

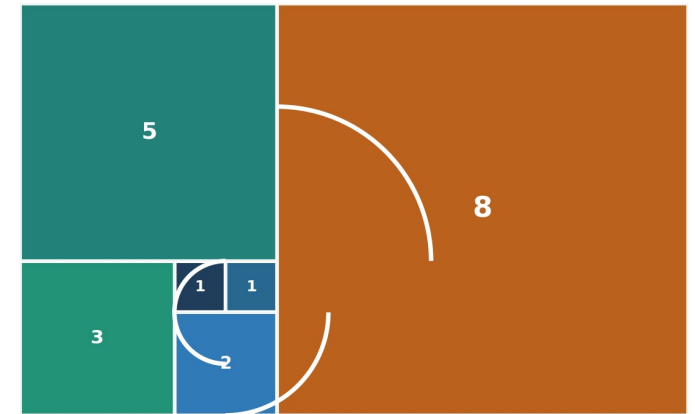


Story Point Estimation

Fibonacci estimation — story points grow non-linearly (1, 2, 3, 5, 8, 13, 21...) so larger, riskier stories are separated clearly from small ones.

User Story	Points	Justification
Registration	3	Standard form and validation.
Login	2	Simple, well-understood auth.
Station Monitoring	5	Live data integration needed.
Charger Availability	5	Real-time status per charger.
Reservation	8	Concurrency and hold logic.
Waiting Time Prediction	8	AI model and data pipeline.
Notifications	5	Event triggers across channels.

Fibonacci Sequence



Each number is the sum of the two before it — gaps widen as uncertainty grows.



Sprint Goal & Deliverables

Sprint Goal Develop user authentication and real-time charger monitoring.



Registration Module



Login Module



Charging Dashboard



Charger Availability



Database Integration



Basic UI



Unit Testing

Milestone Timeline — 2-week sprint

Day 1–2
Setup & DB

Day 3–5
Auth modules

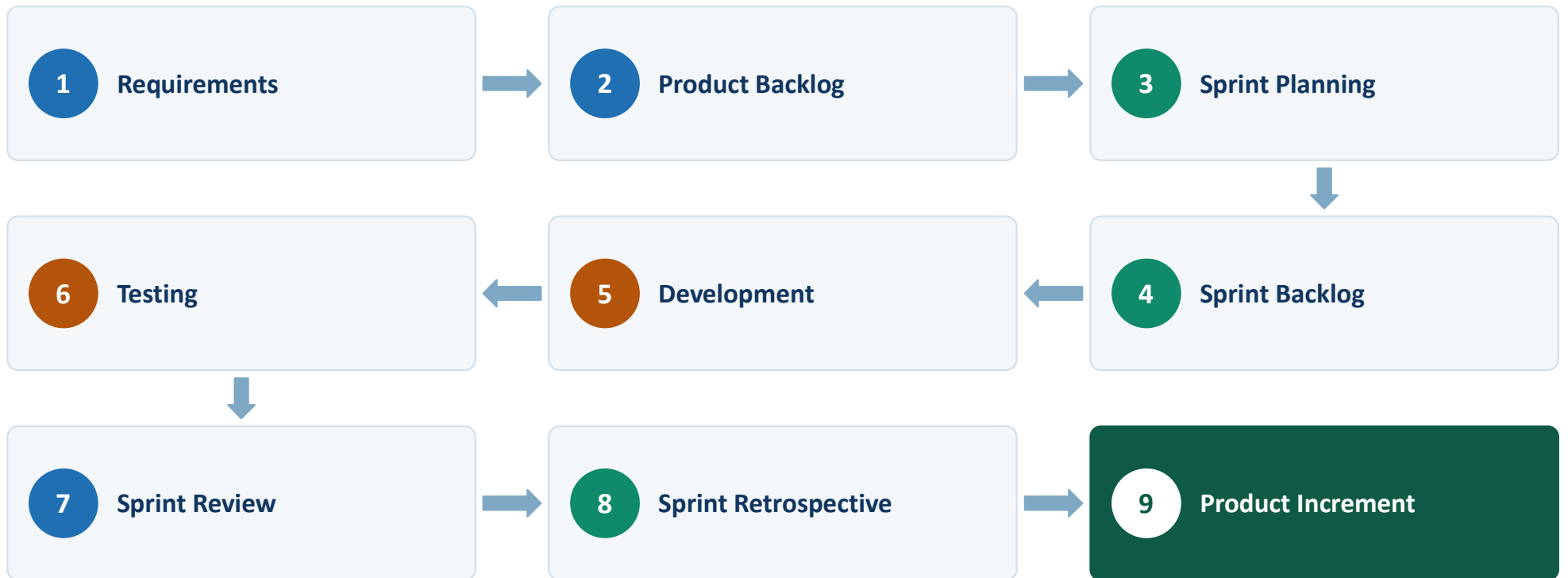
Day 6–9
Monitoring

Day 10–12
UI & integrate

Day 13–14
Test & review

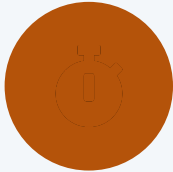


Agile Workflow



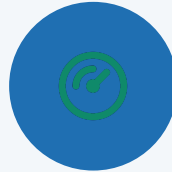


Expected Outcomes



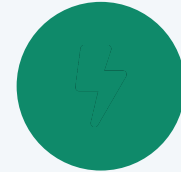
Reduced waiting time

Reservations and predictions cut queueing at busy stations.



Better charger utilisation

Smart allocation balances load across the network.



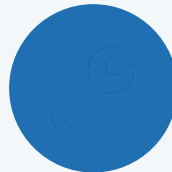
Improved energy management

Consumption tracking and reports optimise electricity use.



Higher customer satisfaction

Reliable visibility and shorter waits improve the experience.



Efficient maintenance

Fault alerts enable faster repair and less downtime.



Smart charging ecosystem

A connected, data-driven, sustainable charging network.

Conclusion



Agile improves project management by delivering value in short, adaptive cycles.



Sprint planning helps prioritise the features that matter most, first.



User stories keep the team focused on real customer needs and outcomes.



Smart EV charging reduces waiting time and optimises energy usage across the network.

Thank You